

In [1]: !python -m pip install pandas

Requirement already satisfied: pandas in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (3.0.0)  
 Requirement already satisfied: numpy>=1.26.0 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2.4.2)  
 Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\manishaa\appdata\roaming\python\python311\site-packages (from pandas) (2.9.0.post0)  
 Requirement already satisfied: tzdata in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from pandas) (2025.3)  
 Requirement already satisfied: six>=1.5 in c:\users\manishaa\appdata\roaming\python\python311\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is available: 24.0 -> 26.0

[notice] To update, run: python.exe -m pip install --upgrade pip

In [2]: !python -m pip install scikit-learn

Requirement already satisfied: scikit-learn in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (1.8.0)  
 Requirement already satisfied: numpy>=1.24.1 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from scikit-learn) (2.4.2)  
 Requirement already satisfied: scipy>=1.10.0 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from scikit-learn) (1.17.0)  
 Requirement already satisfied: joblib>=1.3.0 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from scikit-learn) (1.5.3)  
 Requirement already satisfied: threadpoolctl>=3.2.0 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from scikit-learn) (3.6.0)

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In [3]: !python -m pip install seaborn

## Collecting seaborn

Using cached seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)  
 Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (2.4.2)  
 Requirement already satisfied: pandas>=1.2 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (3.0.0)  
 Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from seaborn) (3.10.8)  
 Requirement already satisfied: contourpy>=1.0.1 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)  
 Requirement already satisfied: cycler>=0.10 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (0.12.1)  
 Requirement already satisfied: fonttools>=4.22.0 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.61.1)  
 Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.9)  
 Requirement already satisfied: packaging>=20.0 in c:\users\manishaa\appdata\roaming\python\python311\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (26.0)  
 Requirement already satisfied: pillow>=8 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (12.1.0)  
 Requirement already satisfied: pyparsing>=3 in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.3.2)  
 Requirement already satisfied: python-dateutil>=2.7 in c:\users\manishaa\appdata\roaming\python\python311\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.9.0.post0)  
 Requirement already satisfied: tzdata in c:\users\manishaa\appdata\local\programs\python\python311\lib\site-packages (from pandas>=1.2->seaborn) (2025.3)  
 Requirement already satisfied: six>=1.5 in c:\users\manishaa\appdata\roaming\python\python311\site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->seaborn) (1.17.0)  
 Using cached seaborn-0.13.2-py3-none-any.whl (294 kB)  
 Installing collected packages: seaborn  
 Successfully installed seaborn-0.13.2

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In [4]: `import pandas as pd`

```
import numpy as np
from sklearn.preprocessing import MinMaxScaler, StandardScaler
import seaborn as sns
import matplotlib.pyplot as plt
```

In [5]: `df = pd.read_csv("diabetes.csv")`  
`df.head()`

Out[5]:

|   | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI  | DiabetesPedigreeFun |
|---|-------------|---------|---------------|---------------|---------|------|---------------------|
| 0 | 6           | 148     | 72            | 35            | 0       | 33.6 |                     |
| 1 | 1           | 85      | 66            | 29            | 0       | 26.6 |                     |
| 2 | 8           | 183     | 64            | 0             | 0       | 23.3 |                     |
| 3 | 1           | 89      | 66            | 23            | 94      | 28.1 |                     |
| 4 | 0           | 137     | 40            | 35            | 168     | 43.1 |                     |

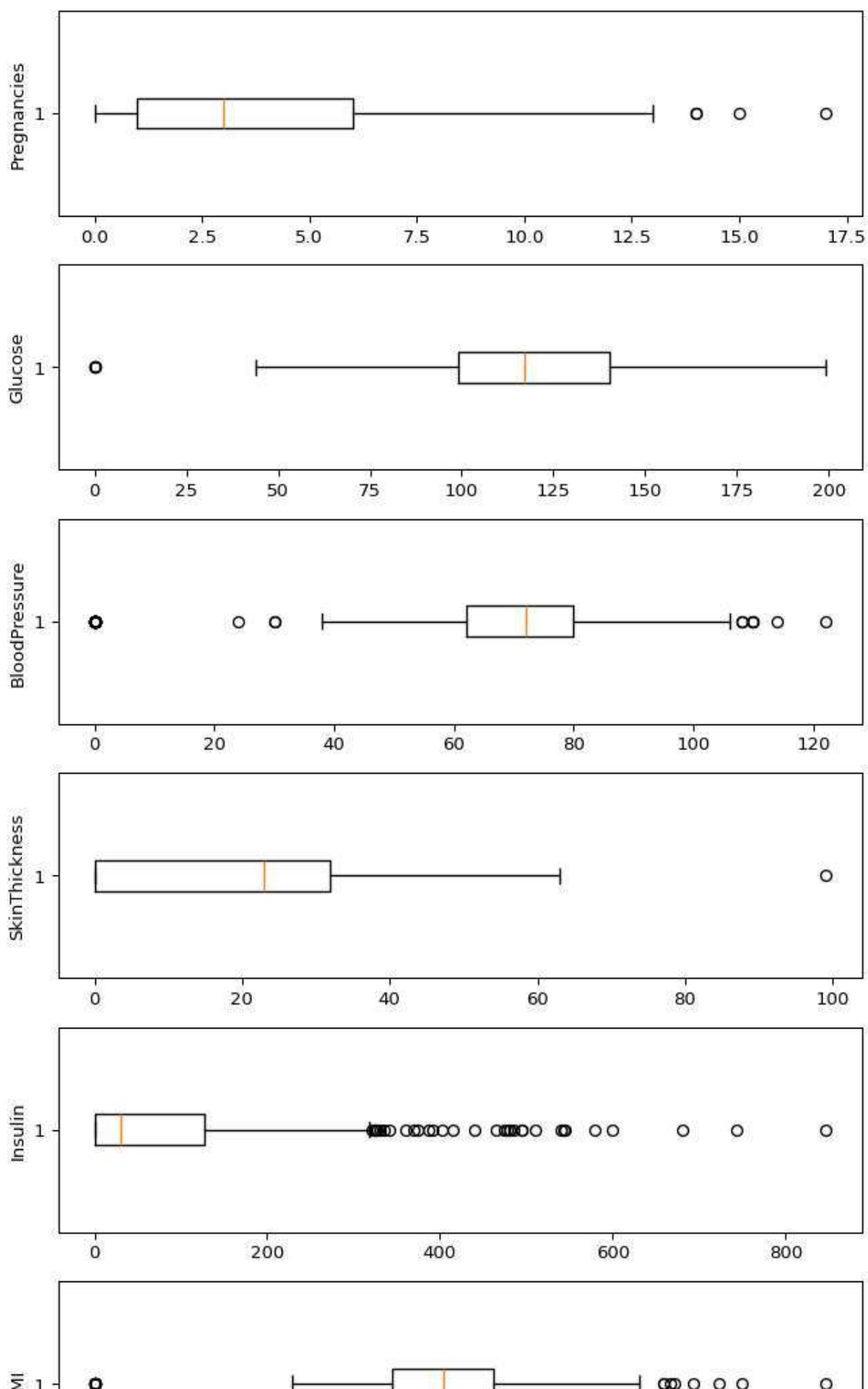


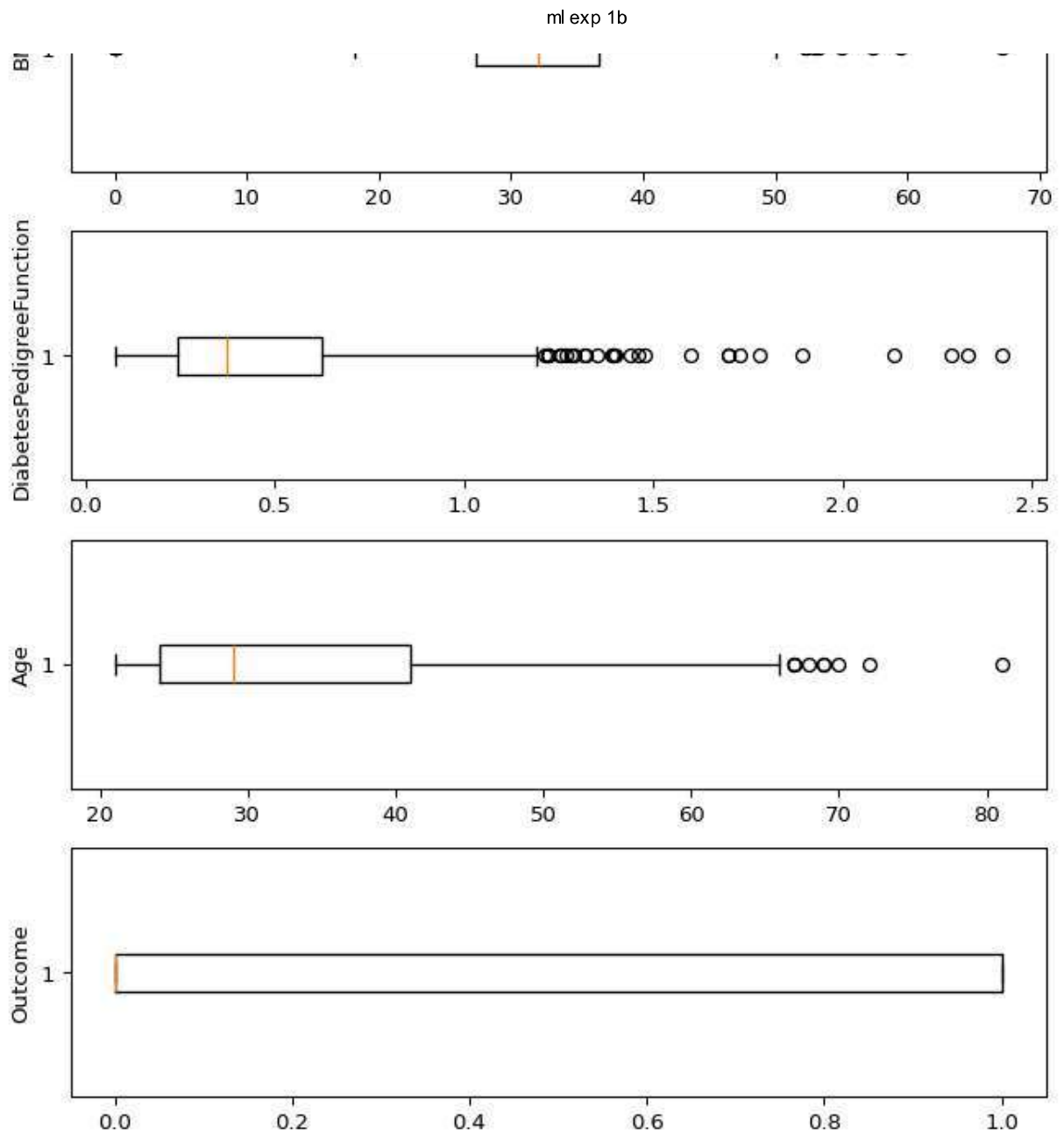
In [6]: `df.info()`  
`print(df.isnull().sum())`

```
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Pregnancies            768 non-null   int64
1   Glucose                768 non-null   int64
2   BloodPressure          768 non-null   int64
3   SkinThickness          768 non-null   int64
4   Insulin                768 non-null   int64
5   BMI                   768 non-null   float64
6   DiabetesPedigreeFunction 768 non-null   float64
7   Age                   768 non-null   int64
8   Outcome                768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
Pregnancies            0
Glucose                0
BloodPressure          0
SkinThickness          0
Insulin                0
BMI                   0
DiabetesPedigreeFunction 0
Age                   0
Outcome                0
dtype: int64
```

In [8]: `df.describe()`

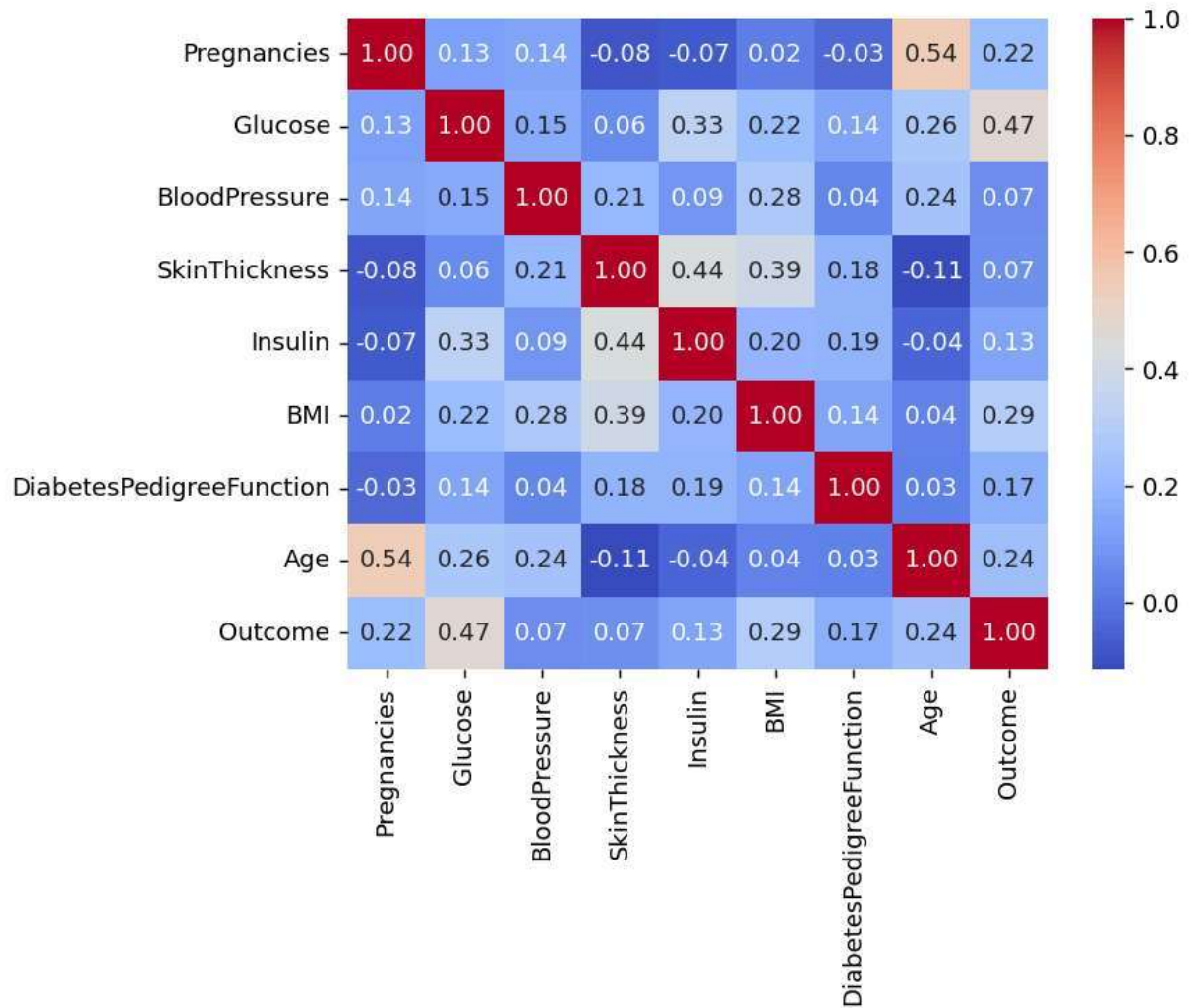
```
fig, axs = plt.subplots(len(df.columns), 1, figsize=(7, 18), dpi=95)
for i, col in enumerate(df.columns):
    axs[i].boxplot(df[col], vert=False)
    axs[i].set_ylabel(col)
plt.tight_layout()
plt.show()
```





```
In [9]: q1, q3 = np.percentile(df['Insulin'], [25, 75])
        iqr = q3 - q1
        lower = q1 - 1.5 * iqr
        upper = q3 + 1.5 * iqr
        clean_df = df[(df['Insulin'] >= lower) & (df['Insulin'] <= upper)]
```

```
In [10]: corr = df.corr()
        plt.figure(dpi=130)
        sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm')
        plt.show()
        print(corr['Outcome'].sort_values(ascending=False))
```



```

Outcome          1.000000
Glucose          0.466581
BMI              0.292695
Age              0.238356
Pregnancies      0.221898
DiabetesPedigreeFunction 0.173844
Insulin          0.130548
SkinThickness    0.074752
BloodPressure    0.065068
Name: Outcome, dtype: float64

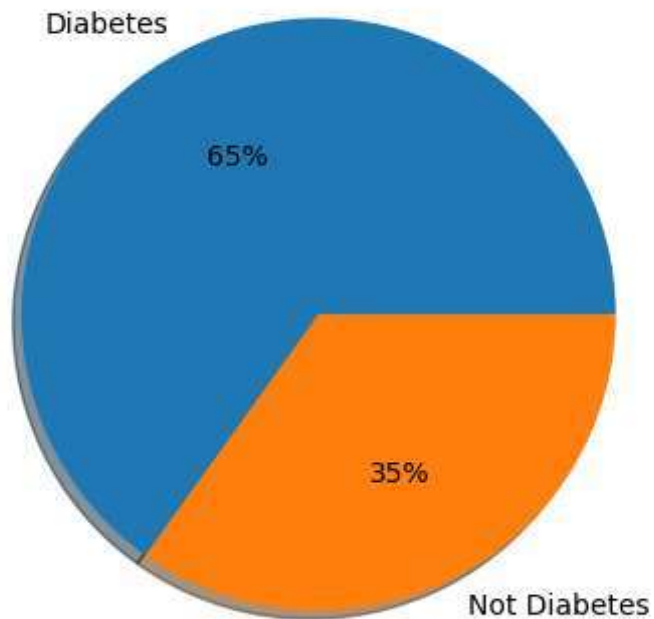
```

```

In [11]: plt.pie(df['Outcome'].value_counts(), labels=[
        'Diabetes', 'Not Diabetes'], autopct='%f%%', shadow=True)
plt.title('Outcome Proportionality')
plt.show()

```

## Outcome Proportionality



```
In [12]: X = df.drop(columns=['Outcome'])
         y = df['Outcome']
```

```
In [13]: scaler = MinMaxScaler()
         X_normalized = scaler.fit_transform(X)
         print(X_normalized[:5])
```

```
[[0.35294118 0.74371859 0.59016393 0.35353535 0.          0.50074516
  0.23441503 0.48333333]
 [0.05882353 0.42713568 0.54098361 0.29292929 0.          0.39642325
  0.11656704 0.16666667]
 [0.47058824 0.91959799 0.52459016 0.          0.          0.34724292
  0.25362938 0.18333333]
 [0.05882353 0.44723618 0.54098361 0.23232323 0.11111111 0.41877794
  0.03800171 0.          ]
 [0.          0.68844221 0.32786885 0.35353535 0.19858156 0.64232489
  0.94363792 0.2          ]]
```

```
In [ ]:
```